

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

**Duration : 45 Minutes
Total Marks:20**

Annexure A

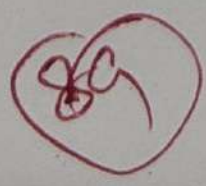
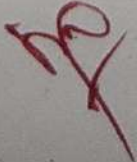
Descriptive Written Test

Code of Conduct:

I Navin Kumar MA (192511015) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Navin Kumar MA

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

CO1- AT3

Assessment

Smart Home Automation System

Introduction

A smart home automation system controls lighting, temperature and security using sensor data such as motion, temperature and time a day. It also learns user preferences to improve comfort and energy efficiency.

Types of Intelligent Agents:-

1. Simple Reflex Agent

- Acts based on current sensor input only.
- Ex: Turns on light when motion is detected.
- Advantages: Fast and simple.
- Disadvantages: Cannot remember previous events

2. Model - Based Reflex Agent

- Maintains an internal model of the environment.
- Ex: Keeps light ON while a person is inside a room.
- Advantages: Works in partially observable environments.
- Disadvantages: More complex than a simple reflex agent.

3. Goal - Based Agent

- Takes actions to achieve a specific goal.
- Ex: Maintains room temperature at 24°C while reducing energy usage.
- Advantages: Makes better decisions.
- Disadvantages: Requires planning.

4. Utility - Based Agent.

chooses the action that gives maximum overall benefit.

• Ex: Balances comfort, electricity cost and security.

• Advantages: Finds the best solution among many choices.

• Disadvantages: Utility calculation is complex.

Best choice :-

• A hybrid agent (Model-Based Utility-Based) is the most effective because it remembers past information, learns user preferences, and selects actions that maximize comfort, security and energy efficiency.

Conclusion:-

• A hybrid intelligent agent provides reliable, adaptive and efficient home automation.

2.

Smart Traffic Management System.

Introduction:-

A smart traffic management system controls traffic lights using real-time data such as vehicle density, weather conditions and pedestrian movement. It also learns traffic patterns over time.

Types of Intelligent Agents:-

1. Simple Reflex Agent

- changes signals based only on current traffic.

- Advantages: Fast response.

- Disadvantages: Cannot learn from previous traffic.

2.

Model - Based Agent.

Maintains traffic history and road conditions.

Advantages:- Better decision-making.

Disadvantages:- Requires more computation.

3. Goal - Based Agent:-

Balances goal to reduce congestion and waiting time.

Advantages:- Optimizes traffic flow.

Disadvantages:- Needs planning.

4. Utility - Based Agent

Balances travel time, fuel consumption, emergency vehicles and safety

Advantages:- Produces the best overall traffic management.

Disadvantages:- Utility calculation is complex

Best Agents:-

A Hybrid Agent (Model-Based Utility-Based) is the most suitable because learns traffic patterns, adapts to changing conditions and optimizes signal timing for all road users.

Conclusion:-

A Hybrid Intelligent agent improves traffic flow, reduces congestion, minimizes fuel consumption and gives priority to emergency vehicles making the transportation system more efficient and safer.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand